Quadratic Data Intelligence

Product Catalog, Q2 2026

Behavioral Intelligence Products for Financial Institutions

Quadratic OÜ · Confidential — for prospective partners only

Executive Summary

The Problem We Solve

Banks understand their clients through transactions: what they bought, when, and for how much. But banks cannot see *why* clients make the decisions they do. They cannot observe the knowledge state at the moment of decision. They cannot measure whether their advisory and education efforts actually change client behavior. The gap between transaction data and behavioral context is a €2B+ annual cost in client churn, regulatory penalties, and missed advisory fees.

Our Unique Advantage

Quadratic captures complete behavioral context by controlling three environments simultaneously: financial education (what users learn), market simulation (how they trade), and social dynamics (what influences them). Every action is recorded with a full knowledge snapshot — quiz scores, portfolio state, bias severity, information exposure, and ECFL certification level — at the exact moment the user trades, invests, or saves. This is the only dataset in the world that answers: **"What did this client know when they made this decision?"**

The ECFL Framework & Certification

Our data is further enriched by integration with the EU Financial Competence Framework (ECFL), which categorizes financial competency into six bands (F1–F6) with rigorous, standardized certification. Every user on our platform can be assigned an ECFL band based on their demonstrated knowledge and trading behavior. This means our behavioral intelligence is stratified by certified competency level — we can show exactly how F1-certified users behave differently from F4-certified users in the same market conditions.

Data Volume & Scaling Projection

io	Monthly Observations	Annual Users	Cohorts Available
.00 users)	10,000	100	5–8
(500 users)	50,000	500	12–15
5,000 users)	500,000	5,000	40–50
(50,000 users)	5M+	50,000	100+

Each observation includes a full context snapshot. With 500 active users, we generate ~50,000 rich observations per month. At 5,000 users, this scales to 500,000/month — sufficient for segment-level statistical analysis across 40+ demographic/behavioral cohorts.

Platform Overview

How the Data Is Generated

The Observation Pipeline

Quadratic generates behavioral observations through a three-layer system:

- **Gameplay Layer:** Users engage in financial trading simulation, education courses, and social competition. Every action (trade, quiz submission, article read, peer follow, market call) generates an event.
- **Context Snapshot:** At each event, a complete knowledge snapshot is recorded: quiz scores, portfolio metrics, bias severity scores, ECFL band, recent information exposure, and game progression state.
- **Warehouse & Analysis:** Observations flow to cloud storage (Supabase) for analysis and product generation. No raw data is ever shared with B2B partners — only aggregated, anonymized, k-anonymized outputs.

The Knowledge Snapshot Model

The core intellectual property is the "knowledge snapshot" — a JSON object capturing the user's complete information state at the moment of decision. Every observation includes:

- **Knowledge:** Courses completed, quiz scores, time on platform, concept mastery per topic
- **Portfolio:** Net worth, cash balance, position count, diversification score, Sharpe ratio
- **Biases:** Severity scores (0–100) for 10 cognitive biases (disposition effect, overconfidence, loss aversion, etc.)
- **Game State:** Level, tier, progression axes (Growth/Governance/Impact/Power), corporate structure
- **ECFL Data:** Certified band (F1–F6), tier classification, domain strengths, certification dates
- **Exposure:** Last articles read, social feed items seen, peer trade alerts, headlines viewed
- **Demographics:** Country, age range (binned), university, registration date

This snapshot answers the question: "What did this user know when they made this decision?" No financial dataset has this combination.

The ECFL Framework

Band	Name	Typical Profile	Example Role
F1	Financial Basics	Identifies income vs. expense	Retail employee
F2	Basic Competency	Understands budgeting, simple products	Mid-career saver
F3	Intermediate	Analyses bonds, basic portfolio theory	Junior analyst

F4	Advanced	Understands derivatives, factor models	Senior trader
F5	Expert	Models correlation structure, volatility	Quantitative analyst
F6	Master	Comprehends systemic risk, macro linkages	Fund manager

Data Privacy & Compliance

GDPR Compliance Framework

All data collection is GDPR-compliant:

- **Consent:** Users opt-in to data collection via granular consent modal. Opt-in only — no pre-checked boxes.
- **Transparency:** Privacy policy details what data is collected, how it's used, and who might receive it.
- **Data Minimization:** Only data necessary for gameplay or product generation is collected.
- **Retention:** Raw observations are retained for 24 months. Users can export their data or request deletion at any time.
- **Data Transfers:** All EU data remains in EU data centres (GDPR-compliant Supabase infrastructure).

Three-Layer Anonymization Methodology

Layer 1: Pseudonymization

- User IDs replaced with deterministic pseudonyms (salted hash)
- Session IDs hashed; IP addresses not stored

Layer 2: K-Anonymity

- Every quasi-identifier combination (country + age range + university) has ≥5 records
- If a demographic cell has < 5 records, it is generalized (France → Western Europe)
- Standard $k \geq 5$ for research datasets, $k \geq 10$ for commercial products

Layer 3: Differential Privacy (Optional)

- Aggregate queries add calibrated Laplace noise
- Mathematical guarantee: privacy loss of ε (epsilon) per query
- Example: "Average Sharpe ratio for F3 French users" — true value ±0.05 with 90% confidence, $\varepsilon = 1.0$

Data Processing Agreement (DPA)

Every B2B partner signs Quadratic's standard DPA specifying: what data they receive (cohort IDs, never individuals); how they can use it (research, business intelligence, regulatory reporting only); retention and deletion timelines; prohibition on re-identification attempts; and annual audit rights.

Product Catalog

10 Data Products for Financial Institutions

#	Product Name	Primary Buyer	Price Range	Freshness
1	Bank Client Intelligence Report	Retail/Private Banks	€15K–€50K/yr	Quarterly
2	Behavioral Finance Research Dataset	Universities, Quant Funds	€5K–€20K/yr	Quarterly
3	Knowledge-Decision Correlation Report	Banks, Regulators	€25K–€100K/yr	Quarterly
4	FOMO Susceptibility Index	Regulators, Insurers	€12K–€25K/yr	Quarterly
5	Cross-Country Financial Behavior Atlas	ECB, Central Banks	€40K/yr	Annual
6	AI Training Datasets	Quant Funds, AI Labs	€8K–€25K/yr	Monthly
7	Real-Time Behavioral Signals API	Algo Trading Firms	€15K–€100K/yr	Hourly
8	Competency-Segmented Intelligence	Wealth Managers	€60K–€80K/yr	Quarterly
9	Certification Funnel Analytics	Corp Training Depts	€20K–€35K/yr	Quarterly
10	Pre/Post Cert Behavioral Shift Report	Banks, Regulators	€30K–€80K/yr	Quarterly

PRODUCT 1

Bank Client Intelligence Report

FLAGSHIP

DESCRIPTION

A quarterly behavioral profile of your client demographic, captured through their trading simulation activity on the Quadratic platform. Shows how your clients actually behave with money — not what they tell their advisor, but what they demonstrate in a realistic market simulation.

BUYER PROFILE

- Retail banks
- Private banks
- Wealth management firms
- Corporate treasury divisions

USE CASE

Your clients spend €50K+ annually on financial advisory and education. But you have no systematic measurement of whether they're learning, improving their risk discipline, or actually changing their behavior. This product shows: your F2-certified retail clients hold 34% more diversified portfolios than F1 clients. Your clients who completed the risk management course reduced their disposition effect score by an average of 23 points.

SAMPLE OUTPUT (JSON)

Sample JSON structure (abbreviated):


```
{
  "reportId": "BIR-2026-Q2-BANK001",
  "cohort": { "bank": "SocGen Private Banking", "country": "France",
              "sampleSize": 847, "observationCount": 68420 },
  "ecflProfile": { "avgBand": 3.4,
    "distribution": { "F1": 8.2, "F2": 24.1, "F3": 42.7, "F4": 18.3 },
    "certificationRate": 67.4 },
  "behavioralProfile": {
    "riskLiteracy": { "avgScore": 0.62, "trend": "+0.08 vs Q1 2026" },
    "dispositionEffect": { "avgSeverity": 58, "range": [12, 94] },
    "diversification": { "avgHerfindahl": 0.34, "percentFullyDiversified": 23.4 },
    "topBiases": [
      { "bias": "Disposition Effect", "avgSeverity": 58 },
      { "bias": "Overconfidence", "avgSeverity": 51 },
      { "bias": "Herding", "avgSeverity": 44 }
    ]
  },
  "tradingBehavior": {
```

```
    "avgTradesPerMonth": 12.3,
    "sharpeRatio": { "median": 0.84, "mean": 0.91 }
  },
  "recommendedInterventions": [
    { "intervention": "Disposition Effect Coaching",
      "target": "Clients with severity > 70",
      "expectedBiasReduction": 0.22, "targetSize": 189 }
  ]
}
```


Data Freshness: Quarterly (90-day rolling window)

PRICING

Base: €15,000/year — 1 cohort, annual updates

Standard: €35,000/year — 1–3 cohorts, quarterly, export formats

Premium: €50,000/year — unlimited cohorts, quarterly, custom analyses, API

PRIVACY GUARANTEES

✓ All data aggregated to cohort level (k ≥ 10)

✓ No individual identification possible

✓ Minimum 50 users per cohort

✓ GDPR DPA included; data retention limited to 24 months

TYPICAL OUTCOMES

- 15–30% improvement in client advisory effectiveness
- 40% better targeting of educational interventions
- Regulatory evidence for MiFID II suitability compliance filing

PRODUCT 2

Behavioral Finance Research Dataset

RESEARCH

DESCRIPTION

Anonymized, k-anonymized dataset of trading decisions, knowledge snapshots, and outcomes. Designed for academic researchers, quantitative firms, and behavioral finance practitioners who need real-world data on how people actually invest.

BUYER PROFILE

- University finance departments
- Behavioral economics research labs
- Quant hedge funds
- Fintech research teams
- Business schools (MBA programs)

USE CASE

Every published behavioral finance paper relies on historical data or laboratory experiments (small samples, artificial conditions). This dataset provides something rare: real trading decisions by real humans in realistic market conditions, with complete information-state context. Academic papers based on this data are publishable in top journals (Journal of Finance, Review of Financial Studies).

SAMPLE CSV SCHEMA

CSV schema — 1,000+ rows per export:

```
observationId, userId_hashed, timestamp, action, asset,
  quantity, price, knowledge_quizScore, knowledge_coursesCompleted,
  portfolio_netWorth, portfolio_diversification,
  biasScore_disposition, biasScore_overconfidence, biasScore_herding,
  ecfl_band, ecfl_certifiedF3, lastArticleReadTime, marketRegime, eventClass

OBS-2026-089234, user_ae8f, 2026-06-15T14:23:00Z, BUY, AAPL,
  100, 189.34, 0.73, 5, 187450, 0.42, 42, 67, 38, F3, 1, 5min, Growth, normal

OBS-2026-089235, user_3c9d, 2026-06-15T14:24:15Z, SELL, TSLA,
  50, 245.67, 0.58, 2, 42310, 0.18, 78, 82, 61, F2, 0, 15min, Volatility, herding_FOMO
```

Data Freshness: Quarterly (quarterly releases with 6-month lag for privacy)

PRICING

Academic: €5,000/year — cohort-level, non-commercial use only

Research: €12,000/year — country-level, publication use, credit required

Commercial: €20,000+/year — ECFL band-level, full dataset, custom granularity

PRIVACY GUARANTEES

✓ K-anonymity ≥ 5 on all quasi-identifiers

✓ Differential privacy applied ($\varepsilon = 1.0$)

✓ Minimum 6-month data lag before release

✓ All users have given explicit consent; license prohibits re-identification

PRODUCT 3

Knowledge-Decision Correlation Report

FLAGSHIP

DESCRIPTION

The statistical relationship between financial literacy (measured by ECFL band) and investment quality (measured by risk-adjusted returns, diversification, realization rates). Does financial education actually change investment behavior? This product answers it with statistical rigor.

BUYER PROFILE

- Banks (strongest value)
- Regulators (MiFID II evidence)
- Education policy makers
- University MBA programs
- Corporate training departments

USE CASE

Banks invest millions in client education. But they have no statistical evidence that this investment improves outcomes. This product provides exactly that: "F3-certified users (post-certification) achieve 40% better Sharpe ratios than F1-certified users, controlling for demographics, experience, and risk tolerance ($p < 0.001$, $n = 2{,}847$)." This is the slide that closes every bank partnership conversation.

SAMPLE STATISTICAL OUTPUT (JSON)

Sample output (statistical findings):


```
{
  "reportId": "KDC-2026-Q2-001",
  "analysisScope": { "observations": 487234, "uniqueUsers": 18420,
    "countries": ["France", "Germany", "Netherlands", "Belgium", "Luxembourg"] },
  "executiveSummary": {
    "headline": "Financial certification demonstrably improves investment outcomes",
    "tagline": "F3-certified users achieve +40% Sharpe improvement and -0.18 disposition effect",
    "statisticalSignificance": "p < 0.001 (95% CI: [+0.31, +0.51])" },
  "primaryFindings": [
    { "metric": "Sharpe Ratio",
      "F2_uncertified": { "mean": 0.64, "n": 4230 },
      "F2_certified":   { "mean": 0.89, "effectSize": 0.31, "p": "<0.001" },
      "F3_certified":   { "mean": 1.52, "effectSize": 0.57, "p": "<0.001" } },
    { "metric": "Disposition Effect (pre/post cert)",
      "preCertMean": 62.3, "postCertMean": 44.2,
      "causalEffect": -16.7, "pValue": "<0.001" },
```

```
    { "metric": "Herfindahl Diversification by Band",
      "F1_certified": 0.34, "F2_certified": 0.24,
      "F3_certified": 0.18, "F4_certified": 0.12 }
  ],
  "regression": { "iv": "isCertified", "coef": 0.41, "p": "<0.001",
                  "R2": 0.34, "interpretation":
    "Certification is associated with +0.41 Sharpe improvement, controlling for demographics" }
}
```

Data Freshness: Quarterly (updated every 90 days with new cohort)

PRICING

Single report: €25,000 — one country, one ECFL band level, fixed period

Annual subscription: €70,000 — 4 quarterly reports, all countries, live dashboards

Premium: €100,000+ — custom analyses, high-dimensional segmentation, consulting

PRIVACY GUARANTEES

✓ All results aggregated to band level (minimum n=30)

✓ No individual or institution identification

✓ Confidence intervals and p-values reported transparently

✓ Pre-registered analysis design; no p-hacking

TYPICAL OUTCOMES

- MiFID II suitability evidence filed with regulators
- Closes bank partnership conversations with quantitative proof
- Comparable effect size to Vanguard behavioral coaching claims (+0.3–0.4 Sharpe)

PRODUCT 4

FOMO Susceptibility Index

REGULATORY

DESCRIPTION

Per-demographic segment FOMO susceptibility score (0–100). Measures how sensitive each user segment is to social proof, peer influence, and FOMO events. Used by regulators, banks, and insurers to assess risk in specific client populations.

BUYER PROFILE

- Regulators (MiFID II, CSSF)
- Banks (risk assessment, advisory design)
- Insurance companies (behavioral underwriting)
- Algo trading firms (sentiment alpha)

USE CASE

Your retail client base shows FOMO susceptibility of 68 (high). When markets rally, they panic-buy. This product shows: clients with FOMO > 70 experience 2.3× higher drawdowns on average. Regulators can use this to identify populations needing extra suitability protections.

SAMPLE OUTPUT (JSON)

Sample output (demographic breakdown):


```json
{
  "reportId": "FOMO-2026-Q2-001",
  "globalStatistics": { "mean": 52.4, "stdDev": 18.3,
    "percentiles": { "p10": 28, "p50": 52, "p90": 76 } },
  "byDemographic": [
    { "cohort": "France, Age 18-24", "sampleSize": 2340,
      "meanFOMOScore": 68.2, "riskRating": "HIGH" },
    { "cohort": "Germany, Age 35-54", "sampleSize": 1870,
      "meanFOMOScore": 38.1, "riskRating": "LOW" },
    { "cohort": "Netherlands, All Ages", "sampleSize": 890,
      "meanFOMOScore": 45.3, "riskRating": "MODERATE" }
  ],
  "byECFLBand": {
    "F1": { "mean": 72.1 }, "F2": { "mean": 58.4 },
    "F3": { "mean": 42.3 }, "F4": { "mean": 28.1 }
  },
  "correlations": { "fomo_vs_sharpe": -0.34, "p": "<0.001",
    "interpretation": "Higher FOMO correlates with lower returns (-0.34 Sharpe per 10pt F
OMO)" }
}
```

Data Freshness: Quarterly

PRICING

Standard: €12,000/year — aggregate index only

Enhanced: €25,000/year — demographic breakdown, ECFL band stratification, trend analysis

PRIVACY GUARANTEES

✓ Demographic-level aggregation only

✓ Minimum cohort size n=50

✓ No individual scores released

PRODUCT 5

Cross-Country Financial Behavior Atlas

POLICY

DESCRIPTION

Annual report comparing financial behavior, risk literacy, and investment outcomes across EU countries. Answers: how do French retail investors compare to German investors? Do regulatory regimes correlate with behavioral differences?

BUYER PROFILE

- EU regulators (ECB, ESMA)
- Central banks
- Academic institutions
- International policy makers

USE CASE

European regulators want to understand: are Italian savers more risk-averse than Dutch savers? Do MiFID II requirements correlate with better outcomes? This product shows comparative behavioral profiles across all EU countries, stratified by ECFL band, with trend analysis and policy implications.

SAMPLE CROSS-COUNTRY COMPARISON TABLE

Country	Avg Sharpe	Diversification	Disposition Effect	FOMO Score	F1+ Cert %	Avg Band	YoY Trend
Germany	1.04	0.42	48	41	74%	2.9	+3.2%
France	0.89	0.38	56	52	68%	2.7	+2.1%
Netherlands	1.12	0.48	42	38	81%	3.1	+4.1%
Italy	0.61	0.28	68	61	52%	2.3	+1.4%
Spain	0.73	0.32	62	57	61%	2.5	+2.3%
Belgium	0.94	0.40	51	46	71%	2.8	+2.8%
Portugal	0.68	0.30	65	59	58%	2.4	+1.8%

Data Freshness: Annual (December release)

PRICING

€40,000/year — full report + interactive dashboard + policy memo

PRIVACY GUARANTEES

✓ Country-level aggregation, minimum n=200 per country

✓ No individual or institution identification

✓ GDPR DPA required for any sub-national analysis

PRODUCT 6

AI Training Datasets

ML / AI

DESCRIPTION

Curated (context, action, outcome) triplets for training financial decision-making AI models. Each triplet includes: complete user knowledge state (context), trading decision (action), and result (outcome).

BUYER PROFILE

- Quantitative hedge funds
- AI research labs
- Fintech companies
- Machine learning teams

USE CASE

You're training an ML model to predict financial behavior or recommend actions. You need training data that pairs decision context with actual outcomes. This product provides 500K+ training examples of "user with [knowledge snapshot] took [action] resulting in [outcome]". Each example is anonymized and labeled with outcome (Sharpe, PnL, diversification change).

SAMPLE TRIPLET FORMAT (JSON)

Context–Action–Outcome triplet format:

```json
{
  "id": "triplet_0001",
  "context": {
    "knowledgeScore": 0.72, "riskTolerance": 68,
    "dispositionSeverity": 42, "portfolioDiversification": 0.38,
    "recentReturns": [0.02, -0.01, 0.03, 0.01, -0.02],
    "volatilityRegime": "normal", "ecflBand": "F3"
  },
  "action": { "type": "BUY", "asset": "MSFT", "sizePercent": 8, "leverage": 1.0 },
  "outcome_30d": {
    "returnPct": 4.2, "sharpeMeasure": 0.52,
    "sharpeImprovement": 0.08, "biasChangeDisposition": -4
  }
}
```

Data Freshness: Monthly (rolling 12-month window)

PRICING

Research license: €8,000/year — academic / publication use

Commercial license: €25,000/year — proprietary AI model training

PRIVACY GUARANTEES

✓ All user IDs pseudonymized before export

✓ Minimum 6-month data lag

✓ License prohibits re-identification

✓ No raw demographic data (age binned, country-level only)

PRODUCT 7

Real-Time Behavioral Signals API

API

DESCRIPTION

Real-time REST API returning aggregated behavioral signals updated hourly: current FOMO level across user base, panic index, overconfidence indicator, herding metric. Designed for trading firms and sentiment platforms.

BUYER PROFILE

- Algorithmic trading firms
- Sentiment analysis platforms
- Market makers
- Risk managers

USE CASE

Your algo-trading strategy monitors retail sentiment through social media and forums. This API adds a new signal layer: real behavioral metrics from actual trading simulation. Get hourly updates on "retail overconfidence at 72/100 (elevated)" or "FOMO at peak 68/100" — adjust your strategy accordingly.

SAMPLE API RESPONSE (JSON)

Sample GET /signals/current response:


```json
{
  "timestamp": "2026-06-30T15:00:00Z",
  "signals": {
    "fomo_index": 67, "fomo_change_1h": 4, "fomo_direction": "rising",
    "panic_index": 34, "panic_change_1h": -2,
    "overconfidence_index": 61, "herding_metric": 58,
    "dispositionEffect_severity": 51,
    "volatility_regime": "normal", "marketPhase": "growth_phase"
  },
  "by_segment": {
    "F1_users": { "fomo": 78, "panic": 42, "overconfidence": 72 },
    "F3_users": { "fomo": 42, "panic": 18, "overconfidence": 38 }
  },
  "historicalContext": { "fomo_percentile_30d": 72, "fomo_percentile_1y": 65 }
}
```


Data Freshness: Real-time (hourly updates)

PRICING

Starter: €15,000/year — 10 req/min, 90-day history

Standard: €40,000/year — 60 req/min, 1-year history

Premium: €100,000/year — unlimited, real-time webhooks, backtest data

PRIVACY GUARANTEES

✓ All signals are aggregated (cohort-level, minimum n=100)

✓ No individual user data exposed via API

✓ HMAC-SHA256 request signing; TLS 1.3

✓ Annual SOC 2 Type II audit

PRODUCT 8

Competency-Segmented Behavioral Intelligence

PREMIUM

DESCRIPTION

Premium version of Product 1, but stratified by ECFL certified band. Answers: how do your F2 clients behave differently from your F4 clients? Enables tiered advisory design based on demonstrated competency level.

BUYER PROFILE

- Premium banks with tiered advisory services
- Wealth management firms
- Private banking divisions
- Corporate training departments

USE CASE

You want to understand: your F4-certified wealth management clients trade differently than F2-certified self-directed investors. This product segments all behavioral insights by ECFL band, showing: "Your F4 clients achieve 2.3× better Sharpe ratios, hold 5× more diversified portfolios, and reduce disposition effect by 40% vs. F2 clients."

BAND-LEVEL COMPARISON SUMMARY

ECFL Band	Avg Sharpe	Herfindahl (lower=better)	Disposition Effect	FOMO Score	Cert Rate
F1	0.42	0.48	72	72	31%
F2	0.68	0.36	58	58	62%
F3	1.12	0.22	44	42	81%
F4	1.48	0.14	32	28	94%
F5	1.82	0.10	24	18	99%

Data Freshness: Quarterly

PRICING

€60,000–€80,000/year — quarterly updates, band-level dashboards, API access

PRIVACY GUARANTEES

✓ Aggregated to band × demographic cell (k ≥ 10)

✓ Minimum 50 users per band per cohort

✓ No institution or individual identification

PRODUCT 9

Certification Funnel Analytics

EDUCATION

DESCRIPTION

Learning analytics for the ECFL certification pathway: where do learners get stuck? What domains show highest failure rates? What's the optimal sequence of courses to certification?

BUYER PROFILE

- Education policy makers
- Corporate training departments
- Universities
- Compliance / training officers

USE CASE

Your organization mandates financial competency certification for all advisors. You want to know: which modules have the highest drop-off rate? Should Domain 3 (risk assessment) come before Domain 2 (products)? This product shows: "Learners who complete Behavioral Bias module before Portfolio Theory show 28% higher certification pass rates."

SAMPLE OUTPUT (JSON)

Sample certification funnel analytics:


```json
{
  "reportId": "FUNNEL-2026-Q2-001",
  "overallMetrics": {
    "totalStarts": 8420, "completedF1": 6340, "completedF2": 3120,
    "completedF3": 1240, "completedF4": 340,
    "conversionF1": 0.753, "conversionF2": 0.492, "conversionF3": 0.398,
    "avgTimeToF1": "28 days", "avgTimeToF2": "67 days", "avgTimeToF3": "142 days"
  },
  "byDomain": [
    { "domain": "Financial Basics & Products", "passRate": 0.91,
      "avgAttempts": 1.2, "weakestTopic": "Bond pricing" },
    { "domain": "Risk Management", "passRate": 0.73,
      "avgAttempts": 2.1, "weakestTopic": "Value-at-Risk calculation" }
  ],
  "sequencingAnalysis": [
    { "sequence": "Basics → Risk → Portfolio → Behavioral",
      "passRateF3": 0.42, "timeToF3": 156, "efficiency": "average" },
    { "sequence": "Basics → Behavioral → Risk → Portfolio",
      "passRateF3": 0.48, "timeToF3": 134, "efficiency": "high",
```

```
      "recommendation": "Behavioral first reduces anxiety, improves subsequent learning"
}
  ]
}
```


Data Freshness: Quarterly

PRICING

Standard: €20,000/year — funnel metrics, domain pass rates

Enhanced: €35,000/year — sequencing analysis, drop-off attribution, custom cohort

PRIVACY GUARANTEES

✓ Cohort-level aggregation only (k ≥ 10)

✓ No individual learner scores or identities shared

✓ Institution name not disclosed in published research

PRODUCT 10

Pre/Post Certification Behavioral Shift Report

CLOSER

DESCRIPTION

The "closing argument" report: quantitative evidence that financial certification materially improves investment behavior. Answers: does your certification program actually work?

BUYER PROFILE

- Banks (partnership closing argument)
- Regulators (policy justification)
- Educational institutions (impact measurement)
- Compliance departments

USE CASE

"How do I know your certification actually changes client behavior?" Answer this report: "Users who earned F3 certification show: +0.35 Sharpe improvement (p<0.001), –22-point disposition effect reduction, +0.15 diversification improvement. Controls for age, experience, and initial risk tolerance. Equivalent to 180bps annual return improvement."

SAMPLE CAUSAL ANALYSIS (JSON)

Sample pre/post causal analysis:


```json
{
  "reportId": "PREPOST-2026-Q2-001",
  "methodology": {
    "studyDesign": "Prospective cohort with matched control group",
    "treatment": "Completed ECFL F2 or F3 certification exam",
    "matchingVariables": ["age", "experience_years", "initial_sharpe", "bias_scores"],
    "nTreatment": 2340, "nControl": 2340
  },
  "primaryResults": [
    { "outcome": "Sharpe Ratio",
      "preCert": 0.72, "postCert": 1.10, "change": +0.38,
      "controlChange": +0.03, "differenceInDifferences": 0.35,
      "pValue": "<0.001", "ci95": "[0.22, 0.48]" },
    { "outcome": "Disposition Effect (0-100)",
      "preCert": 64, "postCert": 42, "change": -22,
      "controlChange": -1, "differenceInDifferences": -21,
      "pValue": "<0.001" },
    { "outcome": "Herfindahl Diversification",
      "preCert": 0.34, "postCert": 0.18, "change": -0.16,
      "controlChange": -0.01, "differenceInDifferences": -0.15,
```

```
      "pValue": "<0.001" }
  ],
  "robustness": [
    { "test": "Placebo: trading frequency", "result": "No effect (p=0.34). Supports causa
lity." },
    { "test": "RD at cert cutoff (70% vs 69%)", "result": "Sharp discontinuity at F3/F4 b
oundary (p<0.001)" }
  ]
}
```


Data Freshness: Quarterly (updated as new certification cohorts complete post-period observation)

PRICING

Single report: €30,000 — one cohort, one certification period

Quarterly subscription: €80,000/year — rolling 4-report analysis, always current

Custom: €50,000+ — by institution, by region, by competency domain

PRIVACY GUARANTEES

✓ All results aggregated (minimum n=100 per cell)

✓ Difference-in-differences design prevents individual attribution

✓ Pre-registered statistical analysis; p-values corrected for multiple testing

✓ GDPR DPA required; no institution-specific identification in published output

TYPICAL OUTCOMES

- Closes bank partnership conversations with quantitative causal evidence
- MiFID II suitability filing: causal proof that certification improves client outcomes
- "Effect size comparable to Vanguard behavioral coaching; independently replicated"

Pricing & Engagement Models

Subscription Tier Packages

Tier	Products Included	Price/Year	Updates	Support
STANDARD	Products 1–3 (Bank Intelligence, Research Dataset, KD Correlation)	€60K–€80K	Quarterly	Email + annual 1hr consult
PREMIUM	Products 1–7 (+FOMO Index, Cross-Country Atlas, AI Datasets, RT API)	€150K–€200K	Monthly	API access + quarterly consult
ENTERPRISE	All 10 products + custom analyses + co-branded research	€300K+	Real-time	Dedicated account manager

Standalone Product Pricing

Product	Standalone	Annual Subscription
1 — Bank Client Intelligence	€25K/cohort	€40K/year
2 — Behavioral Finance Dataset	€12K	€20K/year
3 — Knowledge-Decision Correlation	€25K	€70K/year
4 — FOMO Susceptibility Index	€12K	€15K/year
5 — Cross-Country Behavior Atlas	€40K	€40K/year (annual)
6 — AI Training Datasets	€8K–€25K	€25K/year
7 — Real-Time Behavioral Signals API	€15K–€100K	Usage-based
8 — Competency-Segmented Intelligence	€60K	€60K/year
9 — Certification Funnel Analytics	€20K	€25K/year
10 — Pre/Post Cert Behavioral Report	€30K	€80K/year

Engagement Models

Annual Subscription

Fixed annual fee. Quarterly or monthly updates. Includes data export rights. 20–30% discount vs. per-report. Recommended for ongoing intelligence use.

Per-Report

One-time purchase of a specific report. No ongoing commitment. Higher per-unit cost. Used for evaluation / proof-of-concept.

API / Metered Access

Pay-as-you-go for Real-Time API. Per-query cost (€0.10–€1.00 per query). Useful for platforms needing live behavioral signals.

Research Partnership

Co-branded academic research. Free data access in exchange for publication rights + attribution. Builds platform credibility in top-20 finance journals.

Regulatory Licensing

Central banks and regulators. Custom bulk rates. Unlimited internal use. EUR 50K+ depending on jurisdiction.

Volume Discounts

- 2+ products bundled: 10% discount
- 3+ products bundled: 15% discount
- Annual vs. quarterly commitment: 20% discount
- Multi-year contracts (2+ years): 25% discount

Data Governance & Compliance

GDPR Compliance Architecture

Lawful Basis:

- **Consent (Article 6(1)(a)):** Primary basis. Users opt-in via explicit modal.
- **Legitimate Interest (Article 6(1)(f)):** Secondary basis for aggregated historical analysis.

Consent Management:

- Granular toggles: data collection, anonymized sharing, experiment participation
- Pre-checked boxes: NONE (opt-in default per GDPR EDPB guidance)
- Revocable at any time via account settings
- Immutable consent log stored separately from data

Data Subject Rights (GDPR Articles 15–21)

Article	Right	Implementation
Art. 15	Access	JSON export of all personal observations on request
Art. 16	Rectification	Contact support to correct demographic data
Art. 17	Erasure	Soft delete after 30-day cooling period; automatic in 24 months
Art. 18	Restrict Processing	Pause collection; retain historical data in anonymized form
Art. 20	Portability	CSV/JSON export in standard format within 30 days
Art. 21	Object	Opt-out of experimentation and data sharing at any time

Three-Layer Anonymization Detail

The following pipeline is applied to every B2B export before data leaves Quadratic servers:

```
Layer 1 — Pseudonymization
  User IDs   → Deterministic hash (HMAC-SHA256, per-product salt)
  Session IDs → Hashed
  IP addresses → NOT stored (deleted at collection point)
  Exact age → Binned (e.g., 44 → "35–44")
  Exact location → Country-level only

Layer 2 — K-Anonymity (Recursive Generalization)
  k=5  for research datasets (academic standard)
  k=10 for commercial products
  k=30 for high-sensitivity aggregate queries
```

```
Algorithm: generalize quasi-identifiers until all cells ≥ k

Example:
  Raw:         { country: "France", age: "25-34", university: "Sorbonne" }
  Records: 2    → violates k=5
  Generalized: { country: "France", age: "25-34", university: "*" }
  Records: 8    → satisfies k=5

Layer 3 — Differential Privacy (Optional, for sensitive queries)
  Noise ~ Laplace(μ=0, b=Δf/ε)
  Δf = max change if one record added/removed
  ε  = 1.0 (standard for published research)
  Example: "Avg Sharpe for F3 French users" → true value ±0.01 (Δf=0.01)
```

Data Processing Agreement (DPA) — Key Terms

DPA Clause	Specification
Processing Activities	Business intelligence, research, regulatory compliance only
Data Controller	Quadratic OÜ determines processing purposes
Data Processor	Partner uses data strictly as instructed in DPA
Processing Location	EU (Ireland Supabase data centre)
Encryption in Transit	TLS 1.3 (AES-256)
Encryption at Rest	AES-256 (Supabase infrastructure)
Access Controls	Role-based (RBAC); audit log of all exports
Security Audit	Annual SOC 2 Type II
Active Retention	24 months
Backup Retention	90 days post-deletion
Re-identification	Contractually prohibited; breach triggers DPA termination
Sub-Processors	Supabase (DB), AWS (backup), Stripe (billing only)
Breach Notification	Within 72 hours per GDPR Article 33
Audit Rights	Partner may audit Quadratic annually (30-day notice)

Technical Integration

REST API — Behavioral Signals

Authentication

```
Authorization: Bearer {API_KEY}
X-Request-Signature: HMAC-SHA256(payload + secret)
Content-Type: application/json
```

Endpoints

GET /signals/current

Returns current FOMO, panic, overconfidence indices (integer 0–100). Rate limit: 60 req/min (standard). Latency: < 200ms.

GET /signals/historical

Query params: start_date, end_date, granularity (hourly|daily). Returns time series of all signals. Rate limit: 10 req/min. Max range: 1 year.

GET /signals/{segment}

Path param: segment = "F1_users" | "F3_users" | "by_country_XX" | "by_age_range_XX". Returns signals for specific cohort.

POST /webhooks/subscribe

Register webhook for signal updates. Trigger: signal changes > 5 points or hourly cadence. Premium tier only. Payload: JSON signal object + timestamp.

Data Delivery Formats

PDF Reports: Auto-generated via ReportLab. 30–40 pages. Include charts, tables, summary statistics. Branded with customer logo.

CSV / JSON Exports: Compatible with Tableau, Power BI, R, Python (pandas). Data dictionary included as separate file.

Data Warehouse Integration: Nightly batch exports via SFTP, or API push to Snowflake, BigQuery, or Redshift. Contact enterprise@quadratic.io.

Data Refresh Schedule

Product	Frequency	Typical Latency
Bank Client Intelligence	Quarterly	72 hours post-quarter-end
Behavioral Finance Dataset	Monthly	2 weeks (privacy processing)
Knowledge-Decision Correlation	Quarterly	1 week

FOMO Susceptibility Index	Daily	1 hour
Cross-Country Atlas	Annual	4 weeks (December release)
AI Training Datasets	Monthly	2 weeks
Real-Time Behavioral Signals API	Hourly	Real-time (< 5 min)
Competency-Segmented Intelligence	Quarterly	1 week
Certification Funnel Analytics	Quarterly	1 week
Pre/Post Cert Behavioral Report	Quarterly	1 week

Appendix A — Glossary of Terms

ECFL Band

EU Financial Competency Framework designation. Six levels F1 (financial basics) through F6 (master level). Users certified via exams embedded in the Quadratic platform.

Knowledge Snapshot

Complete information state recorded at the moment of action. Includes: quiz scores, portfolio metrics, bias scores, ECFL band, demographic data, recent information exposure.

Observation

Single behavioral event (trade, quiz, course completion) with attached knowledge snapshot and metadata. The fundamental unit of Quadratic's behavioral dataset.

K-Anonymity

Privacy guarantee. Every quasi-identifier combination appears in ≥k records. Makes re-identification mathematically difficult.

Disposition Effect

Behavioral bias: tendency to sell winning investments quickly and hold losing investments too long. One of the most pervasive retail investor biases.

Overconfidence

Believing one's knowledge or skill is greater than it actually is. Measured by calibration between stated confidence and actual quiz accuracy.

Herding

Following the crowd; buying when others buy, selling when others sell. Measured by latency and size of trades following social feed alerts.

Sharpe Ratio

Risk-adjusted return metric = (return – risk-free rate) / standard deviation. Higher is better. Typical range on Quadratic platform: 0.3–1.8.

Herfindahl Index

Portfolio concentration measure. 1.0 = all in one asset. 0.0 = perfectly diversified. Quadratic uses this as primary diversification metric (lower = better).

Differential Privacy

Mathematical privacy guarantee. Calibrated Laplace noise ensures no single record materially affects query results. ε (epsilon) controls privacy/accuracy tradeoff.

FOMO

Fear of Missing Out. Susceptibility to making decisions based on perceived peer success or market rallies. Measured by behavioral response to social proof stimuli.

GDPR

General Data Protection Regulation (EU Regulation 2016/679). Binding across all EU member states and EEA. Governs all personal data collection and processing on Quadratic.

DPA

Data Processing Agreement. Legal contract specifying how personal data will be processed. Required for every B2B partner relationship under GDPR Article 28.

MiFID II

Markets in Financial Instruments Directive II (EU 2014/65/EU). Article 25 mandates suitability assessment for financial advice. ECFL certification provides supporting evidence.

Difference-in-Differences

Causal inference methodology. Compares change in treatment group vs. change in matched control group. Used in Products 3 and 10 to establish certification causality.